

Chapter 10: Compare and Contrast Security Implications of Different Architecture Models

Securing the Network

- A multi-layered defense using **firewalls**, **ACLs (Access Control Lists)**, **IDS (Intrusion Detection Systems)**/**IPS (Intrusion Prevention Systems)**, and **SIEM (Security Information and Event Management)** is essential for robust protection.
- **Firewalls** with default "deny all" rules and specific **allow lists** ensure traffic control.
- **ACLs** filter traffic based on defined criteria; **IDS/IPS** detect and prevent malicious activity in real-time.
- **SIEM** aggregates logs and alerts from across the network, offering a unified view and faster threat response.

Securing the Servers

- Key servers like domain controllers and SQL servers must be protected as they handle authentication and sensitive data. Mail servers and video conferencing platforms are common attack targets. Cloud storage solutions (e.g., AWS S3, Azure, Google Cloud) offer built-in encryption, access control, and monitoring to enhance security.

Securing the Hosts

- Host security involves protecting endpoints such as user devices with antivirus software, EDR tools, and MDM systems. Multifactor authentication (MFA) strengthens access control by requiring multiple forms of identity verification, reducing the risk of unauthorized access.

Infrastructure as Code (IaC)

- IaC uses scripts (e.g., YAML, JSON) to automate and manage infrastructure, replacing manual processes.
- Enhances **efficiency**, **consistency**, and **scalability** across environments.
- Supports **version control**, team collaboration, and infrastructure reproducibility.
- Tools: **Terraform**, **Ansible**, **Puppet**, **Chef**; supported by AWS, Azure, and Google Cloud. These tools help organizations build scalable, reliable, and efficient IT systems.

Serverless

- Eliminates server management—CSPs (Cloud Service Providers) handle infrastructure, scalability, and security.
- Developers focus on deploying code, while **BaaS (Backend as a Service)** handles backend functions like authentication and storage.
- Offers **cost efficiency** (no hardware investment), but the customer remains responsible for application-level security and data protection.

Internet of Things (IoT)

- IoT includes smart devices with sensors and connectivity, enabling real-time monitoring and automation.
- Security concerns:
 - **Lack of standardization, insecure communication, and data privacy risks.**
 - Vulnerabilities due to **long device lifecycles, physical access, and global supply chains.**
 - **User awareness** and manufacturer responsibility are critical for proper device configuration and maintenance.

SCADA (Supervisory Control and Data Acquisition)

- A type of **Industrial Control System (ICS)** used in industries like energy, manufacturing, and logistics.
- **Figure 10.6** illustrates a 4-level SCADA architecture:
 - **Level 0 (Plant Field Level):** Sensors, actuators.
 - **Level 1 (Controller Level):** PLCs managing real-time control.
 - **Level 2 (Coordinating Computer Level):** HMIs (Human Machine Interface)/Coordinating Computers for centralized monitoring.
 - **Level 3 (Program Logic Controller Level):** Program Logic Controllers overseeing operations.

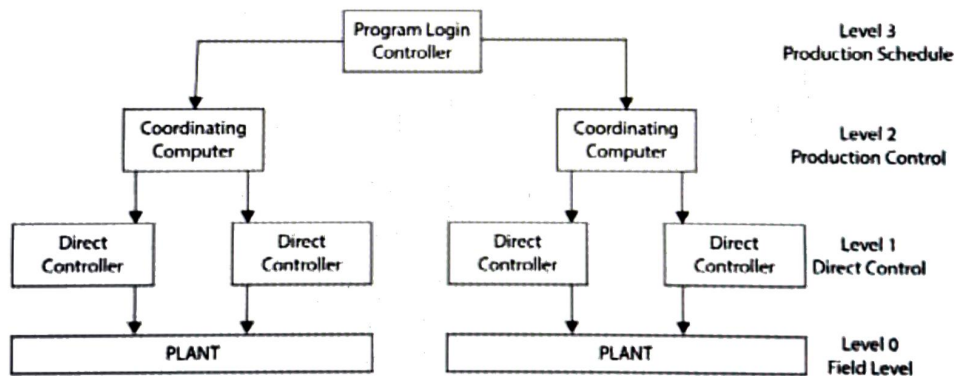


Figure 10.6: SCADA system

- SCADA systems control critical infrastructure sectors like energy, facilities (HVAC), manufacturing, logistics, and industrial processes (e.g., steel production, water treatment), making them essential for smooth operations.
- Due to their vital role, SCADA systems are high-value targets for cyberattacks, requiring robust cybersecurity measures to prevent disruptions in essential services.